higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

WELDING
NQF LEVEL 2

(6031012)

12 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NCV

QUESTION 1

Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–K) next to the question number (1.1–1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.1	Electric shock	A	electrical current delivered by welding power source during welding
1.2	Start voltage	B	welding current flowing through all conductive material
1.3	Start current	C	cable connected from welding power source to electrode holder
1.4	Welding voltage	D	voltage measured across arc
1.5	Welding current	E	average voltage measured at power source
1.6	True arc voltage	F	highest and lowest value of welding current for given voltage load
1.7	Welding circuit	G	state of shock caused by electrical contact
1.8	Welding supply cable	H	voltage between output terminals of a welding power source when there is current
1.9	Welding current range	I	fumes given off when welding
1.10	Load voltage	J	voltage value during starting time
		K	current value during starting time

(10 × 1) **[10]**

QUESTION 2

2.1 Explain how to connect the components for a shielded metal arc welding (SMAW) process. (5)

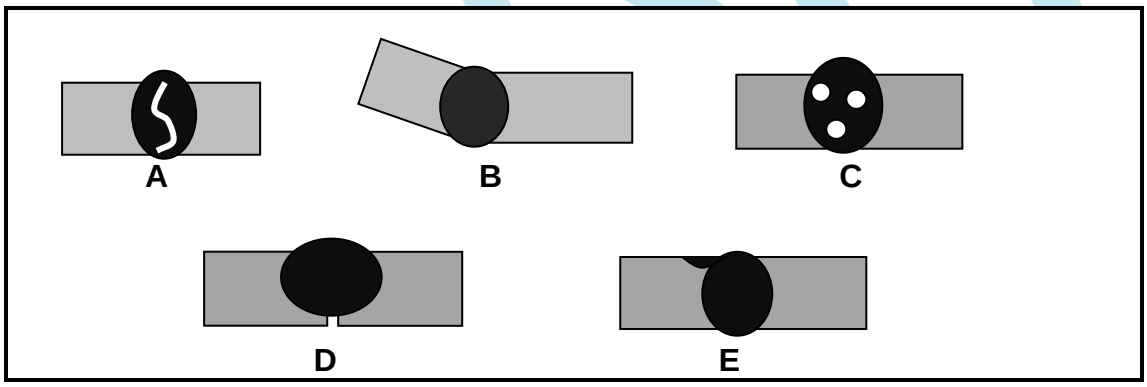
2.2 List FIVE points to consider before welding is done in a certain area. (5)

2.3 Name FOUR ways of storing chemicals. (4)

2.4 Make a neatly labelled sketch of the front and side view of the welding angle of an electrode. Show the direction of the weld and the angle at which the electrode must be held in both views. (6)

2.5 FIGURE 1 shows different types of defects in SMAW.

Identify the defects by writing the answer next to the letter (A–E) in the ANSWER BOOK.



(5)

FIGURE 1

[25]

QUESTION 3

3.1 List FOUR advantages of oxy-acetylene gas welding. (4)

3.2 FIGURE 2 shows an oxy-acetylene gas welding torch.

Identify the parts by writing the answer next to the letter (A–E) in the ANSWER BOOK.

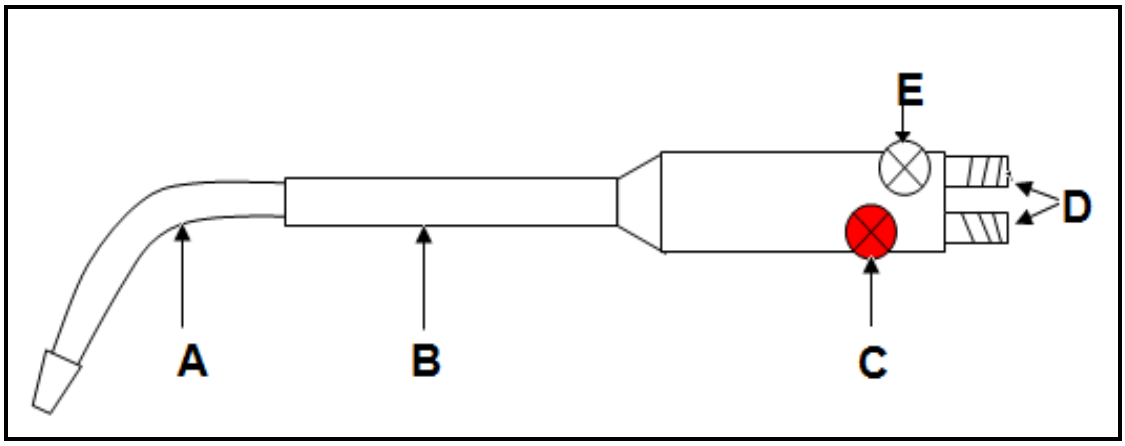


FIGURE 2 (5)

3.3 State which oxy-acetylene equipment has left-hand threads on the connections. (1)

[10]

QUESTION 4

4.1 State the purpose of each of the following components in oxy-acetylene cutting equipment:

4.1.1 Gas cylinder

4.1.2 Gas regulator

4.1.3 Gas hose

4.1.4 Flashback arrestor

4.1.5 Cutting torch

(5 × 2) (10)

4.2 FIGURE 3 shows an artisan preparing a cutting torch to cut a workpiece.

Explain how the artisan sets the torch to a neutral flame.



FIGURE 3 (5)

4.3 The table below shows various cutting pressures, nozzle sizes and thickness of material to be cut.

Copy the table in the ANSWER BOOK and complete it by filling in the missing information next to the letter (A–E).

Plate thickness (mm)	Nozzle size	Oxygen pressure (KPA)	Acetylene pressure (KPA)
12	1	**A**	21
20	**B**	270	**C**
50	**D**	**E**	21

(5)

TABLE

4.4 Explain the importance of wearing leather gloves when cutting. (2)

4.5 List THREE causes of flashback in oxy-acetylene cutting. (3)

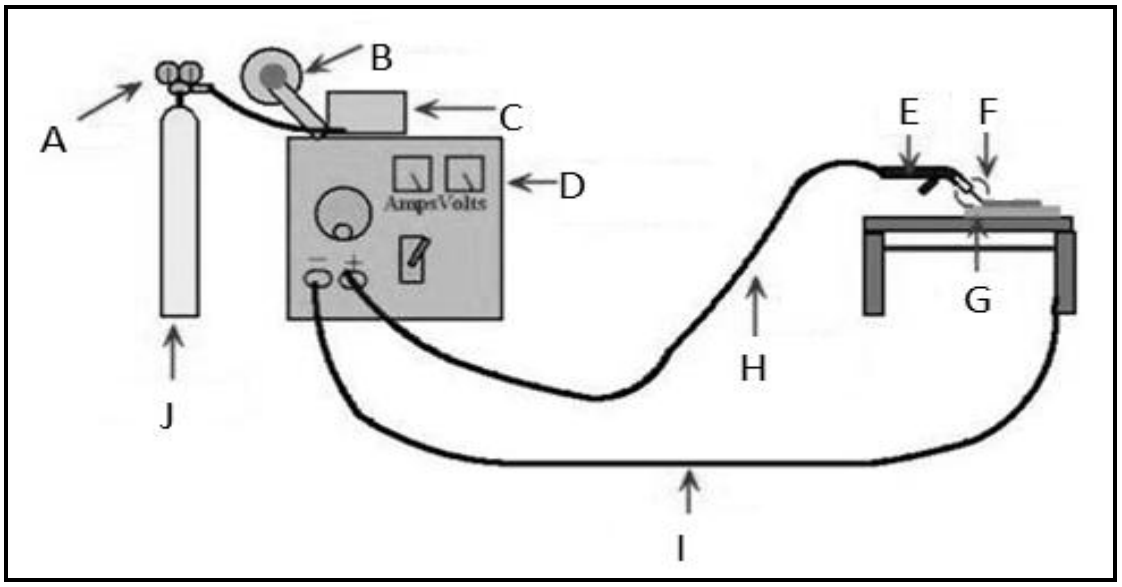
[25]

QUESTION 5

5.1 List FIVE advantages of gas metal arc welding (GMAW). (5)

5.2 Give FIVE pre-operational checks to prepare a safe working area for GMAW. (5)

5.3 Identify the GMAW components in FIGURE 4 by writing the answer next to the letter (A–J) in the ANSWER BOOK.



(10 × 1) (10)

FIGURE 4 **(20)**

QUESTION 6

6.1 List FOUR points to consider when storing acetylene cylinders. (4)

6.2 Make a neat, labelled sketch of the following welding techniques on a butt joint:

6.2.1 Leftward welding technique

6.2.2 Rightward welding technique

(2 × 3) (6)

[10]

TOTAL: 100

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